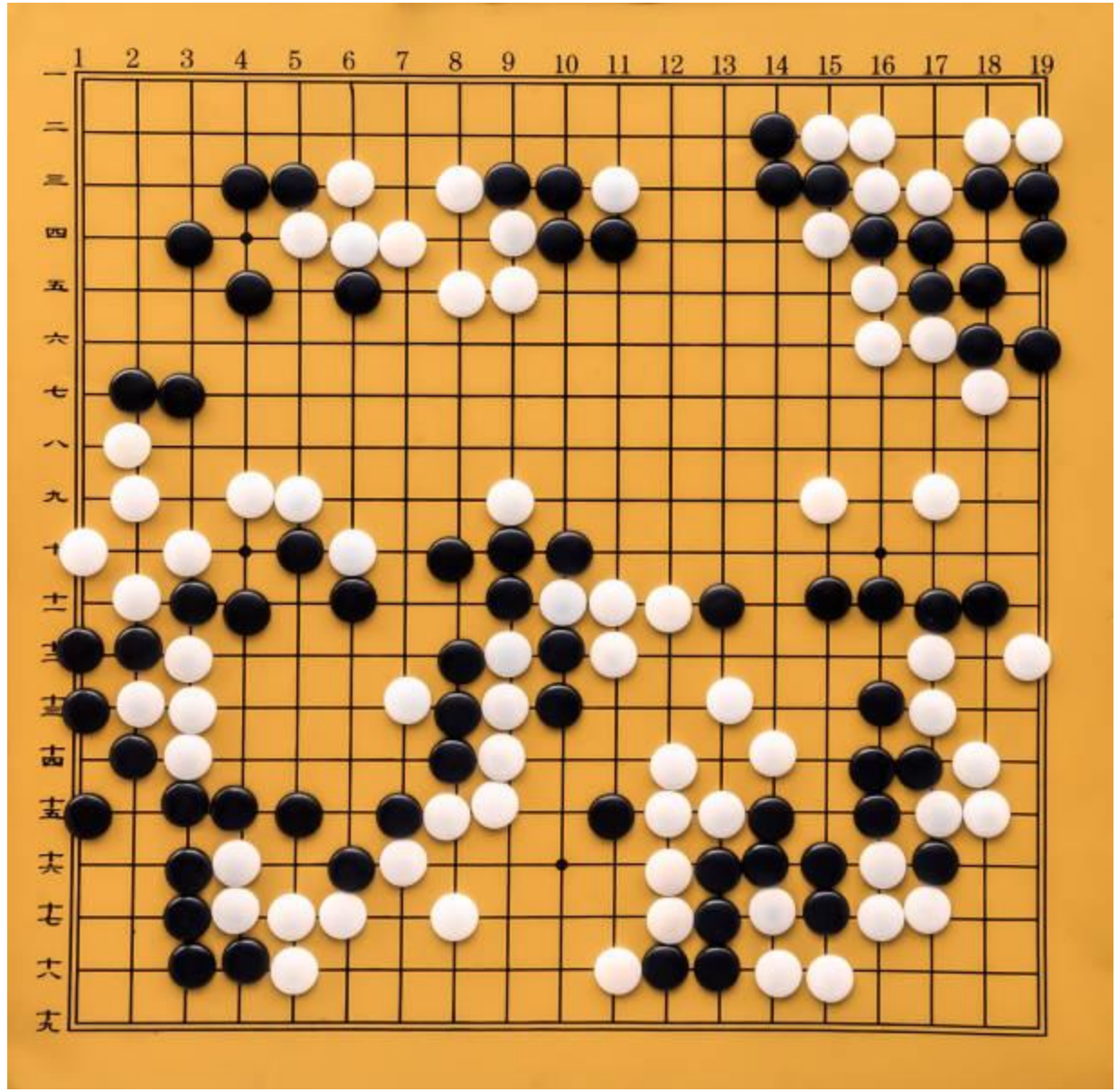
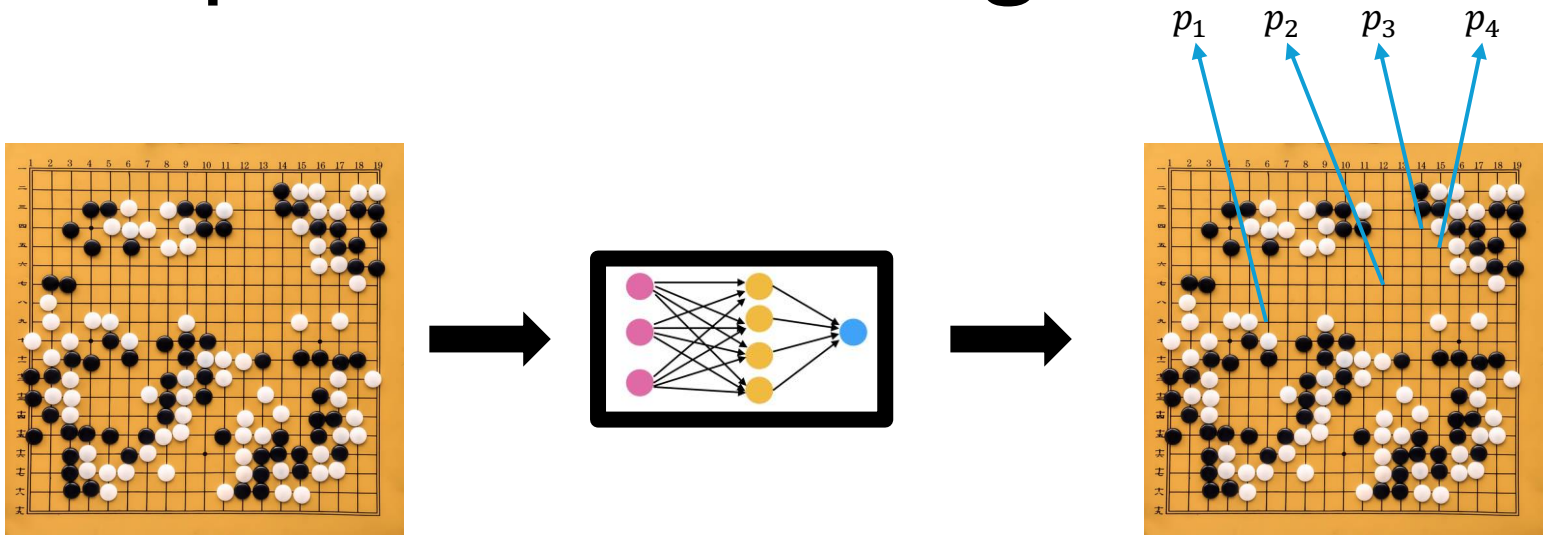
Introduction to
Real-World Reinforcement Learning

Ido Greenberg

2026


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Go with Supervised Learning



- Classification: learn to choose the “right action”
  - Input = board state
  - Output = action probabilities
- Need data of “right actions” (e.g. professional games records)

Supervised Learning: Limitations

- Can’t learn without supervision
- Can’t beat the supervision

2016: AlphaGo uses RL to beat Lee Sedol



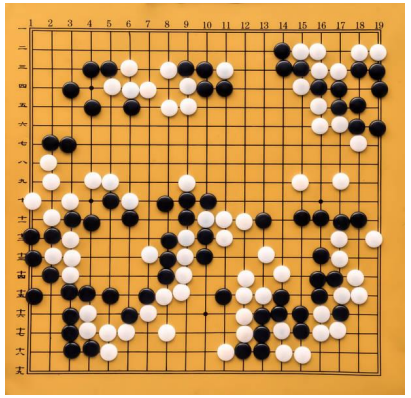
Supervised Learning is not enough

Often we **can't** provide the right choice,
but we **can** measure how good a choice is

Supervised Learning
"You did *A*; you should've done *B*"

Reinforcement Learning
"You did *A*; your score is *R*"

How to Go beyond supervision?



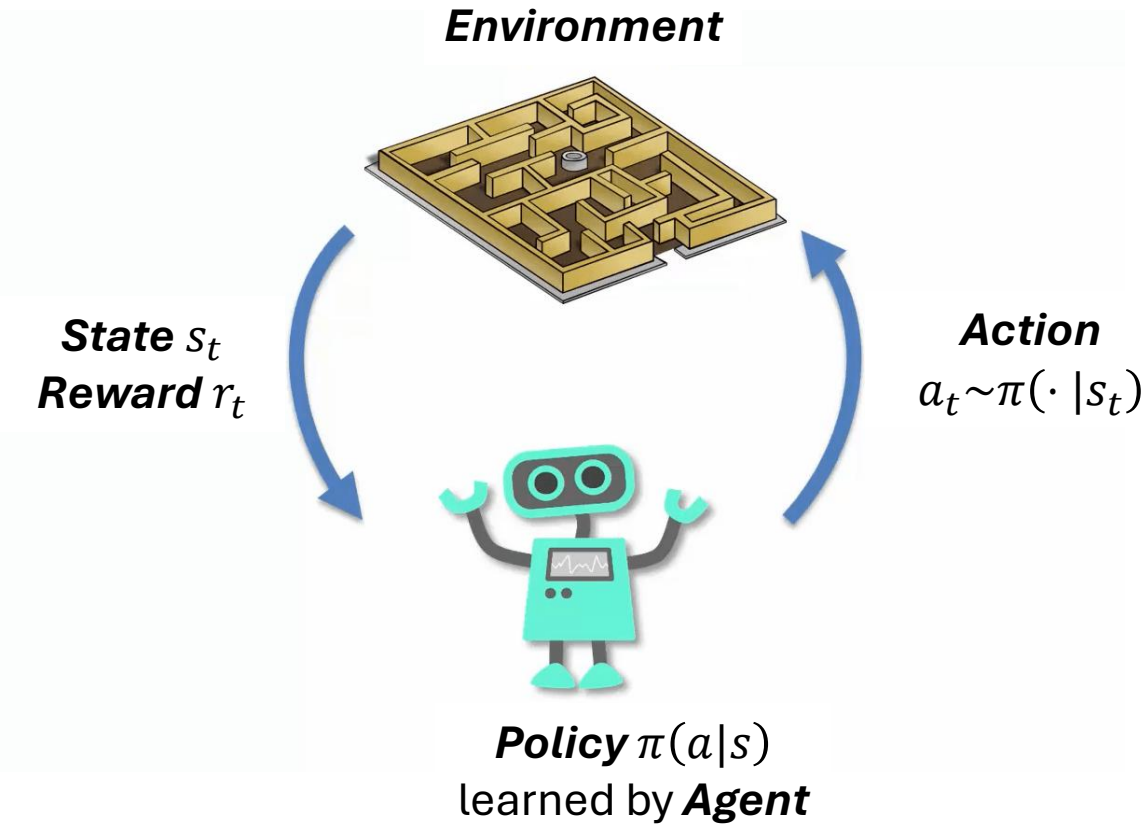
- On training:
  - Which actions to try? (*Exploration*)
  - How to measure success?
  - Which action caused the success?
  - Which opponent actions to simulate?
- On inference:
  - How to use compute to look ahead? (*Planning / Search*)

Reinforcement Learning (RL)

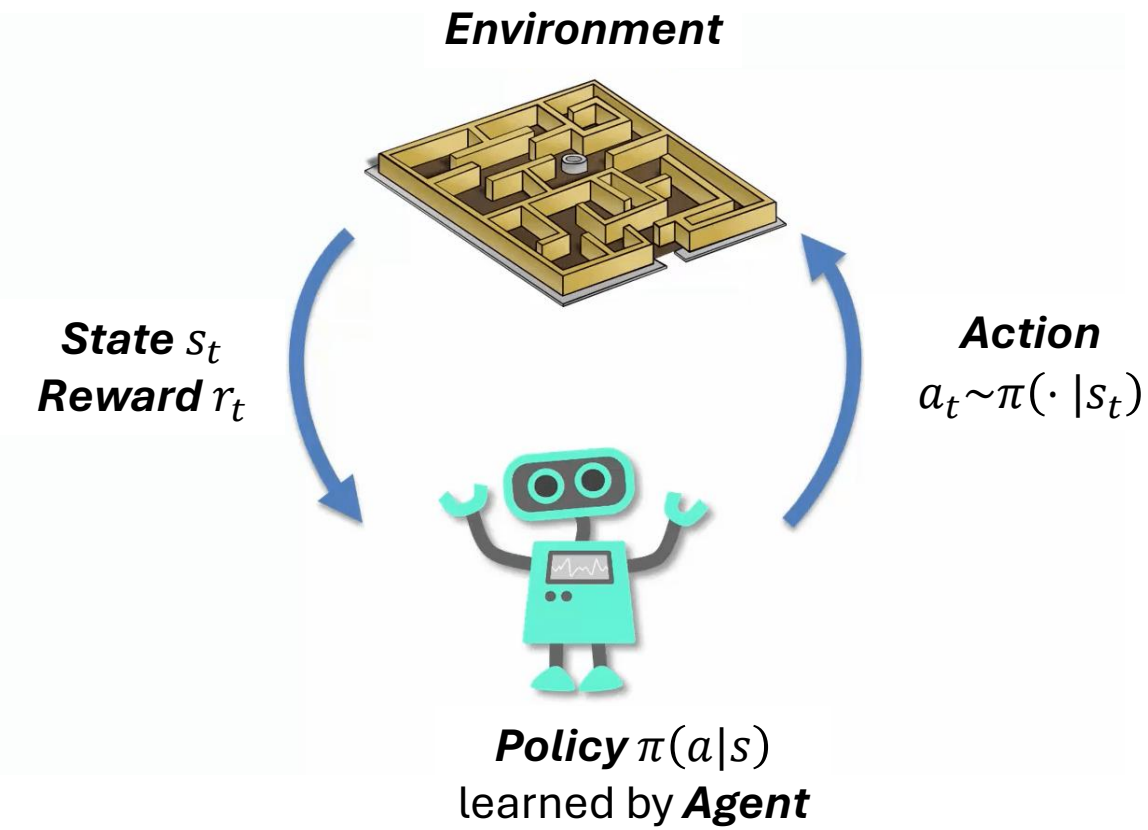
Learn from data / experience

to make decisions under uncertainty

Reinforcement Learning (RL)

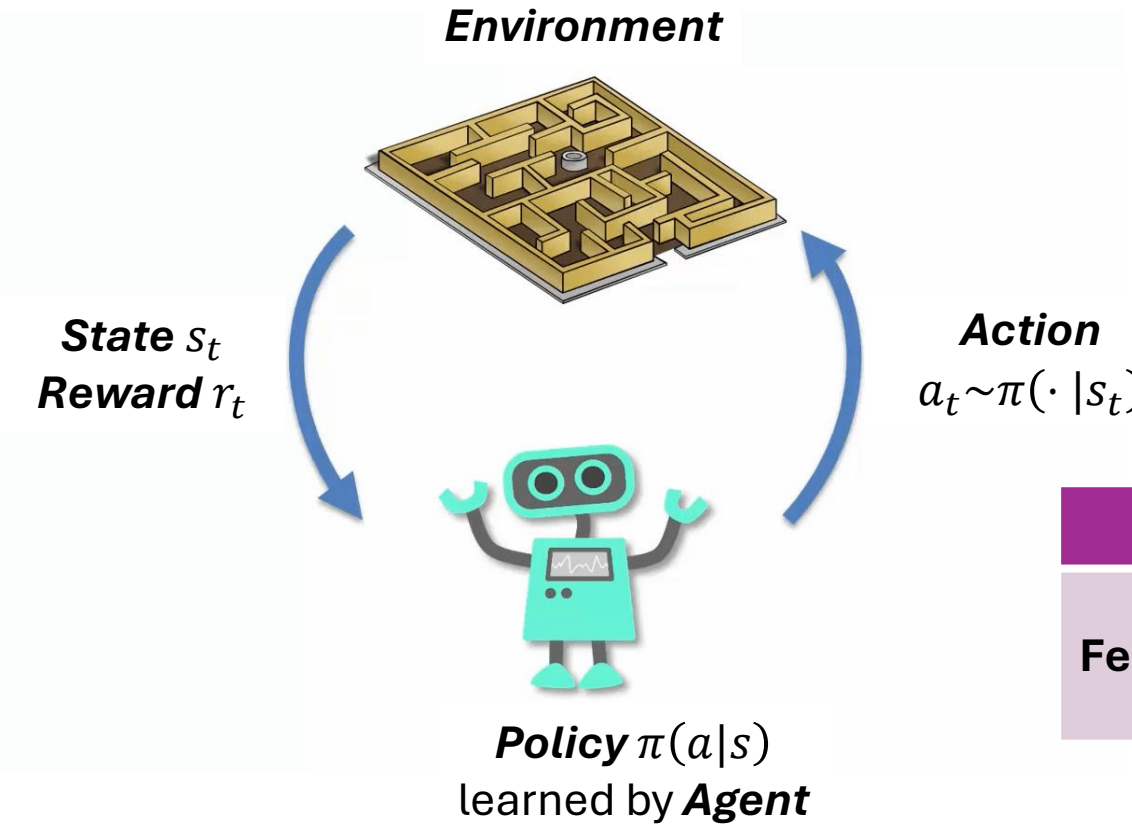


Reinforcement Learning (RL)



- ***Episode***: $\{(s_t, a_t, r_t)\}_{t=0}^{T}$
- ***Return***: $R := \sum_t r_t$
- Goal: $\max_{\pi} E_{\pi}[R]$

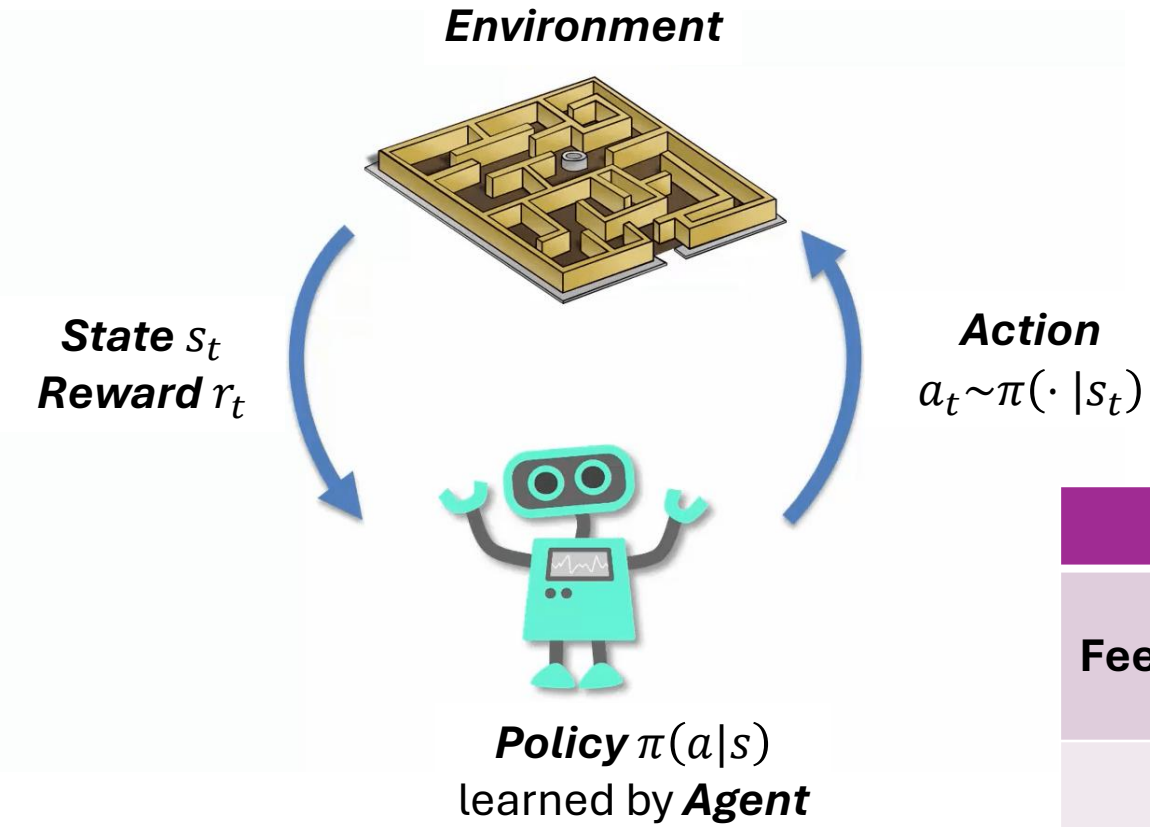
RL vs. Supervised



- ***Episode***: $\{(s_t, a_t, r_t)\}_{t=0}^{T}$
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	Supervised	RL
Feedback	Output you should've chosen	Output's reward

RL vs. Supervised



- ***Episode***: $\{(s_t, a_t, r_t)\}_{t=0}^{T}$
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	Supervised	RL
Feedback	Output you should've chosen	Output's reward
Interactive?	**No:** inputs are independent of previous outputs	**Yes:** output (action) affects next input (state)

RL: some applications



Board games



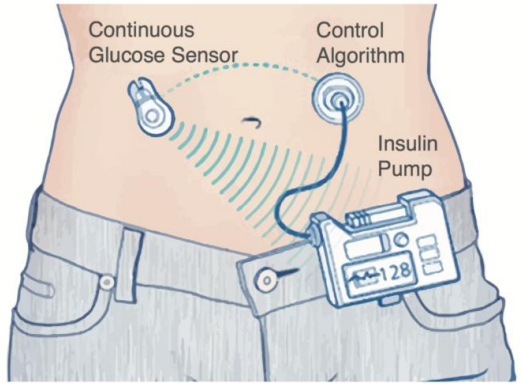
LLMs



Driving



Strategy



Medical devices / treatment



Robotics



Disney Research

Example: RL for LLMs

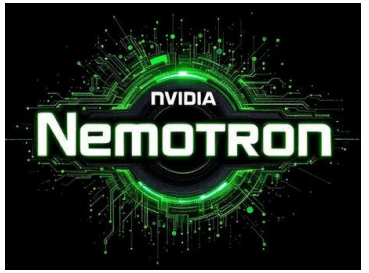
- State = text so far
- Action = next token (word)
- Reward = feedback at the end of answer





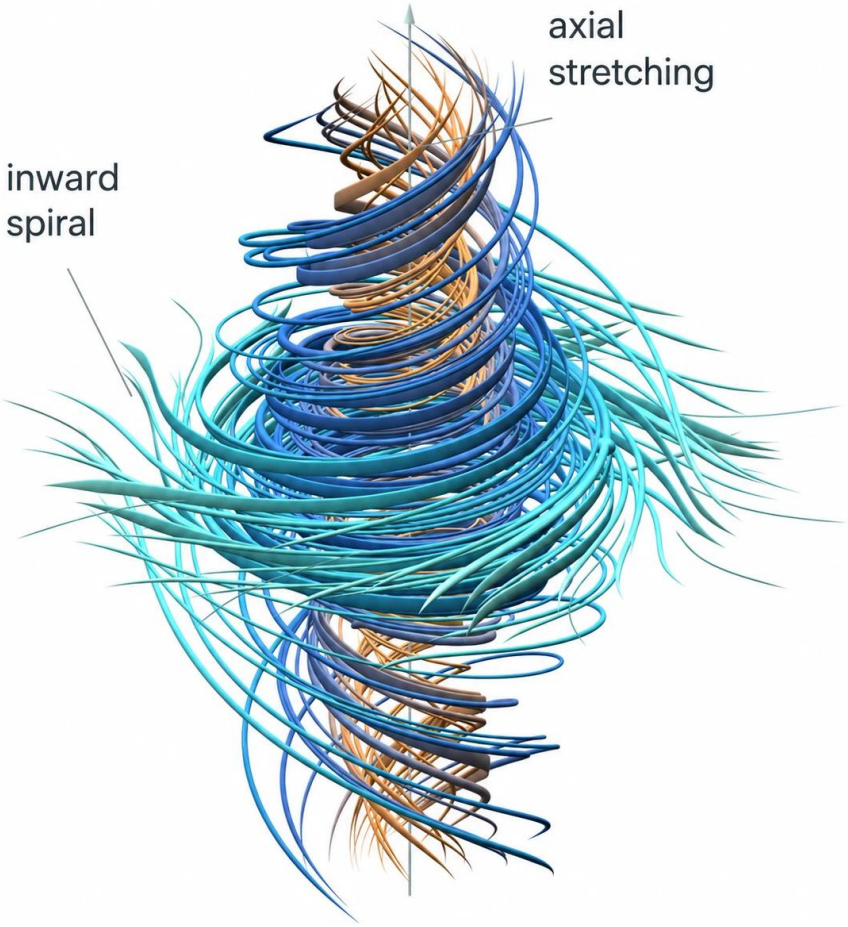






Example: RL for LLMs

- State = text so far
- Action = next token (word)
- Reward = feedback at the end of answer



- Why RL?
  - Go beyond mimic text
  - Alignment with preferences (sentiment / privacy / harmfulness)
  - Distribution shifts (train for different situations from data)
  - Coding & reasoning (new math proofs, reasoning behind the scenes)

Example: RL prevents LLM hallucinations

What is the premise of "Cannonball Run II"? (Num mentions in pretrain corpus: **3**)
Standard SFT: The film begins with a brief recap of the events of the first film, with the death of Jenson (Jerry Reed) and the arrest of Jackie (Burt Reynolds).
RL+Conservative RM: The Cannonball Run II is a cross-country car race.

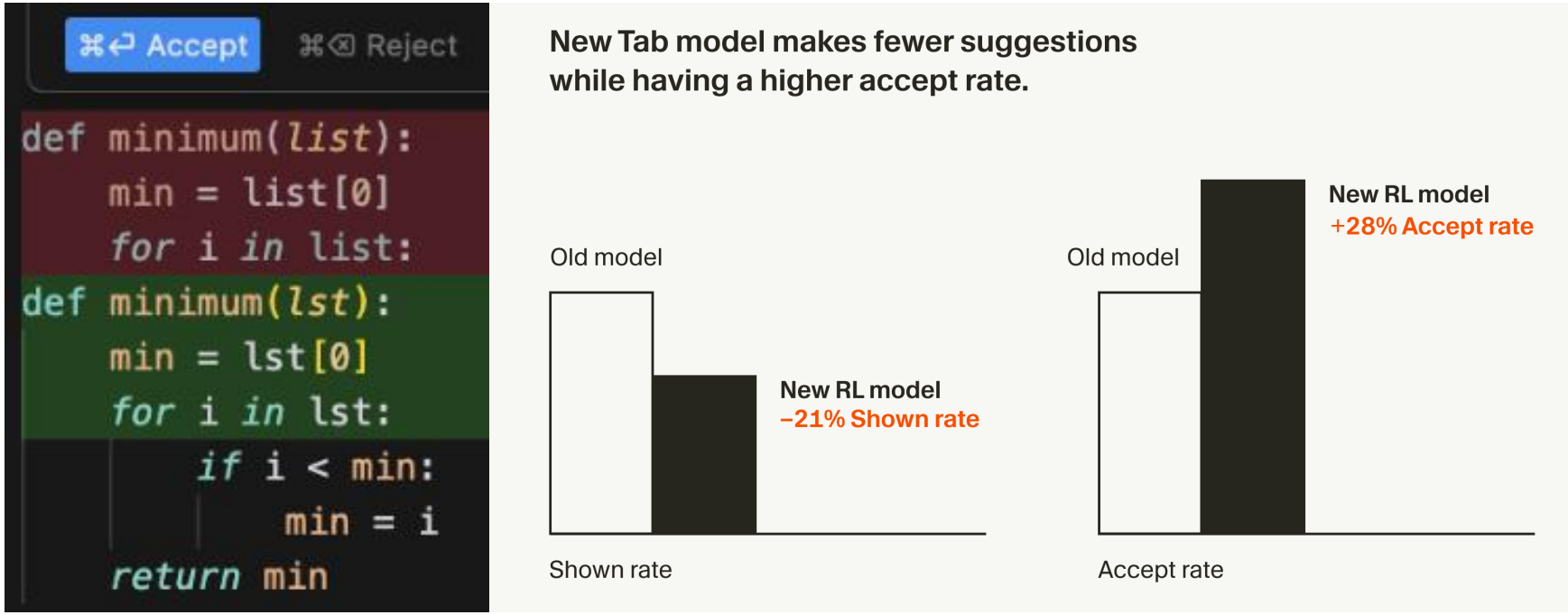
Supervised fine-tuning (SFT)

RL with a conservative reward model

(reward can be adjusted to our needs)

Katie Kang et al. (Berkeley & DeepMind), *Unfamiliar Finetuning Examples Control How Language Models Hallucinate*, 2024

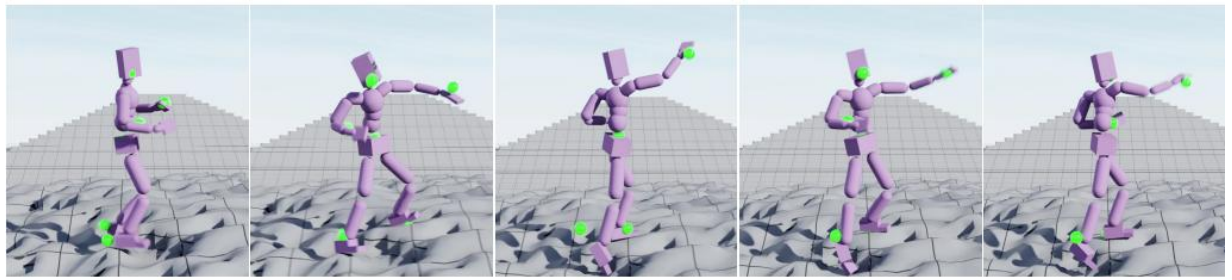
Example: RL improves Cursor's code suggestions



Jacob Jackson et al., *Online RL for Cursor Tab*, Cursor's blog, 2025

When to use RL – even when having supervision

- Example: mimic motion from examples (**supervised**)

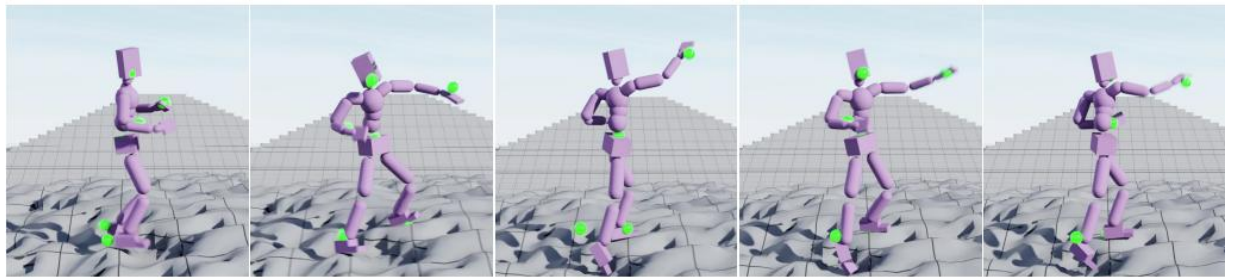


(a) **Full-body tracking:** punching

Tessler et al., *MaskedMimic*, SIGGRAPH 2024

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- Small motion errors → out-of-distribution (e.g. fall)

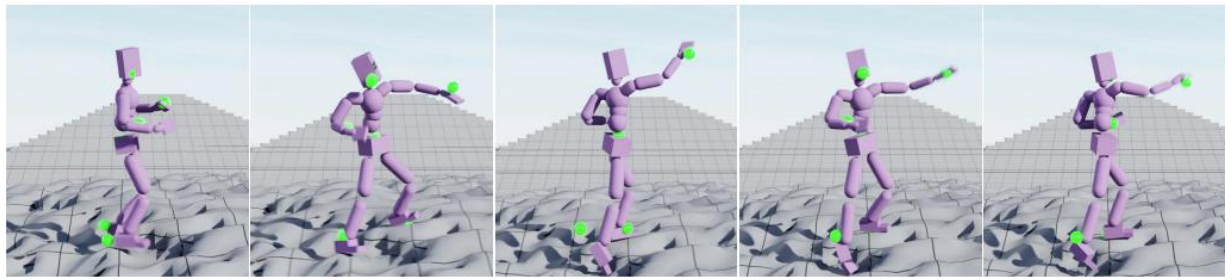


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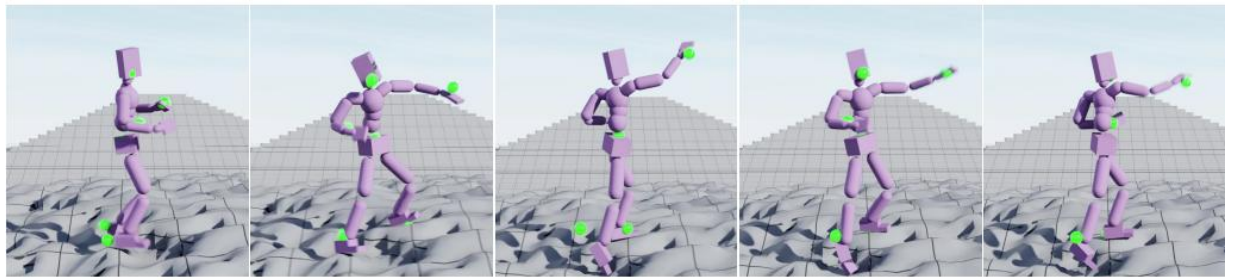


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Tessler et al., *MaskedMimic*, SIGGRAPH 2024

When to use RL – even when having supervision

- Example: mimic motion from examples (**supervised**)
- Small motion errors → out-of-distribution (e.g. fall)
  - **Supervised:** no examples of falling → can't learn to recover
  - **RL:** learn reward → can train beyond original data



(a) **Full-body tracking:** punching

Tessler et al., *MaskedMimic*, SIGGRAPH 2024

When *NOT* to use RL

- True supervision available
  - Example: car plate identification



ABC1234

When *NOT* to use RL

- True supervision available
  - Example: car plate identification



- Policy derived from perception
  - Example: limited driving (e.g. keep distance)



When *NOT* to use RL

- True supervision available
  - Example: car plate identification



- Policy derived from perception
  - Example: limited driving (e.g. keep distance)



- No decision, just prediction
  - Example: Expected Goals (xG)



Summary

- Reinforcement Learning:
  - Feedback = reward
  - Interactive: decision affects next state